Code Quality & Formatting Specification

PRETTIER V3.X

1. Executive Overview & Purpose

To maintain clean code quality, consistency, and smooth pull request workflows across our app, we have integrated **Prettier** as our standard automated code formatter. In a full-stack architecture combining a Node.js/Express backend with a React/TypeScript frontend, differing code styles lead to unnecessary merge conflicts and noisy Git diffs. Prettier enforces a uniform aesthetic across all JavaScript, TypeScript, CSS, JSON, and Markdown files.

2. Configuration & Standards (.prettierrc)

Our code formatting rules are central to the root-level `.prettierrc` configuration file:

Indentation & Lines

`"tabWidth": 4`
Enforces 4-space indentation.

`"printWidth": 80`
Wraps code lines exceeding 80 characters.

Quotes & Commas

`"singleQuote": true`
Uses single quotes for strings.

`"trailingComma": "es5"`
Adds ES5 trailing commas to minimize diffs.

4. NPM CLI Commands

Command	Action & Usage Context
`npm run format`	Automatically formats and overwrites all tracked source files across both frontend and backend.
`npm run format:check`	Audits source files for formatting compliance without altering code (used in CI pipelines)